

TITANIC SURVIVORS PREDICTION

Project Report

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Industrial Training & Presentation (ECS791)

Degree

BACHELOR OF TECHNOLOGY (CSE)

PROJECT GUIDE:

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SUBMITTED BY:

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FACULTY OF ENGINEERING AND COMPUTING SCIENCES

TEERTHANKER MAHAVEER UNIVERSITY, MORADABAD

ACKNOWLEDGEMENT

I have taken many efforts in this project. However, it could not be feasible without the kind help and assistance of many people and groups. I would like to extend my honest gratitude to all of them.

I am distinctly indebted to **Ms. Rohaila Naaz** for her steering and regular supervision in addition to for the presenting necessary statistics concerning the assignment and also for her aid in finishing the project.

I would love to specify my gratitude closer to my parents & members of Faculty of Engineering and Computing Sciences (FOECS), Teerthanker Mahaveer University for their kind co-operation and encouragement which helped me to complete this challenge.

I would really like to specify my unique gratitude and way to enterprise teachers for giving me such attention and time.

My thanking and appreciations additionally attend my colleagues in suggesting the task and folks who have volitionally helped me out with their skills.

Manish Gupta

Place: Teerthanker Mahaveer University, Moradabad

Date: 08/09/2020

DECLARATION

I hereby declare that this Project Report titled “**TITANIC SURVIVORS PREDICTION**” submitted by me and approved by our project guide, to the Faculty of Engineering and Computing Sciences (FOECS), Teerthanker Mahaveer University, Moradabad, is a bonafide work undertaken by me and it is not submitted to any other University or Institution for the award of any degree diploma / certificate or published any time before.

Student Name:	Manish Gupta	Signature
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Project Guide: (External)	Name	Signature
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Project Guide: (Internal)	Ms. Rohaila Naaz	Signature
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Brief About the Company

The institute/organization from where I have completed my training is **Internshala Trainings.**

Internshala is a renowned internship and online training platform, based out of Gurgaon, India. Founded by Sarvesh Agrawal, an IIT Madras alumnus, in 2010, the website helps students find internships with organisations in India.

The platform, which was founded in 2010, started out as a WordPress blog that aggregated internships across India and articles on education, technology and skill gap. Internshala launched its online trainings in 2014. As of 2018, the platform had 3.5 million students and 80,000 companies.

Scholiverse Educare Private Limited (Internshala)'s Corporate Identification Number (CIN) is U80901HR2011PTC064443 and its registration number is 64443.

Organizational Address: Scholiverse Educare Pvt. Ltd. B-610, Unitech Business Zone, Nirvana Country, South City 2, Gurgaon, India – 122018.



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1 Project Title

Titanic Survivors Prediction – Machine Learning

2 Problem Statement

The sinking of the Titanic is one among the foremost infamous shipwrecks in history. While there was some element of luck involved in surviving, it seems some groups of individuals were more likely to survive than others. In this project, my task as a Data Scientist is to build a predictive model that answers the question: “what sorts of people were more likely to survive?” using passenger data (ie name, age, gender, class, etc). The dataset used in this project can be obtained through Kaggle: <https://www.kaggle.com/c/titanic>

Machine Learning is the Art and Science of making the Machines Learn by themselves, it’s a systematic study of Computer Algorithms using Mathematical techniques like Statistics, Modeling and Computational Techniques like programming, it is a subset of Artificial Intelligence. Since we’re still in the phase of development of Narrow AI, ML is one of the hottest trends in Industry as of now and specialists are highly in demand by all Industries, it finds applications in Filtering, Computer Vision, and in NLP in the Linguistics, as Predictive Modelling in the Business Problems and Data Mining in such Industries. Machine Learning can be subdivided into 3 major categories, based on the type of problems the machine is solving and their nature, they are:

- ❖ Supervised Learning – Like Regression and Classification problems, Target Present
- ❖ Unsupervised Learning – Like Data Density and Clustering Problem, Untargeted
- ❖ Reinforcement Learning - Involving feedback based learning, out of experience

And, there are mixed types of learning problems that can be made out of the basic three one’s as mentioned above, however, these are most important one’s out there and the one’s that are most commonly used which form the basis of Machine Learning.

3 Project Description

This project is a simple yet powerful model that builds upon the basis of biggest maritime disasters in the history on the night of 15th of April, 1912 on Royal Mail Ship Titanic in which over 1500 people died and around 700 survived, the soul of the project is inspired from that incident and here, we’ve built a model to simply predict who will be the survivors of that incident which are traveling on RMS Titanic, we are provided with the Input Dataset containing details of all the Individuals on board and based on that, we are going to generate prediction dataset for the Survivors as the output of the Model in our Project. The domain of the Project is Machine Learning, however, since it being an extensive one, just going with Binary Classification problem (Supervised), here for now.

3.1 Scope of the Work

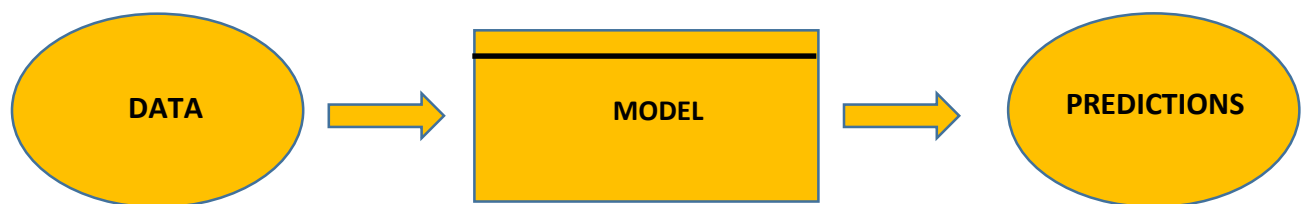
- Can be used cross platform any time subject to IDE availability as it's an IPython Notebook.
- Can Predict Survivors of RMS Titanic with almost 72 percent accuracy in testing phase.
- The resultant predictions are almost instantly exported as another dataset by the model.
- However, unsuitable for Regression and Clustering Problems in the same domain.
- It is extensively used in the industry with manifestations as other business problems.

3.2 Project Modules

This ***Project runs on a Web Browser as an IPython (Jupyter) Notebook*** which can be run on the Systems having the Anaconda IDE Installed, this Project's notebook can then be successfully run on such machines on a Web Browser, calling the jupyter notebook command in CMD and importing the notebook thereby running the model with datasets.

In the Notebook, we can see the use of various Libraries as mentioned are being imported initially, then dataset is loaded and operations on it are performed as per ML Life Cycle and at the end, after building the model, the prediction dataset generated by the model can be seen being exported by the notebook.

3.3 Context Diagram (High Level)



4 Implementation Methodology

Please See Annexures

5 Technologies to be used

5.1 Software Platform

Libraries

- Pandas
- Numpy
- Matplotlib
- Scikit Learn

Back-end

- Python

5.2 Hardware Platform

- a) RAM : at least 4 GB of RAM
- b) Hard Disk : at least 5 GB of Disk Space on Hard Disk
- c) OS : Any Operating System
- d) IDE : Anaconda3 (4.3.1 x64) IDE – Python 3.6.0

5.3 Tools

Tool Name: IPython Notebooks (Jupyter Notebooks) by Anaconda (4.3.1 x64) IDE

Version: 4.3.1 x64 running on Python 3.6.0

Developer(s): Anaconda Inc. (Continuum Analytics Ltd.)

Written in: Python

Browser Used : Google Chrome (Recommended) or Mozilla Firefox

6 Advantages of this Project

- Can be used cross platform any time subject to IDE availability as it's an IPython Notebook and can be enhanced by retraining model on same large dataset.
- Can Predict Survivors of RMS Titanic with almost 72 percent accuracy in testing phase.
- The Same model could be retrained for other similar problems of Binary Classification.
- The resultant predictions are almost instantly exported as another dataset by the model.
- It is extensively used in the industry with manifestations as other business problems.
- The Project doesn't takes any user data or third party data apart from the input dataset for its usage, being concerned and inclined towards user or business privacy without denying user for accessing any feature of the Project.

7 Assumptions, if any

Basically, "Titanic Survivors Prediction" model needs a laptop or a desktop with the Anaconda IDE installed in prior to machine before running it along with the necessary data required to make predictions upon. Thus, the Model Notebook presumes to have a Machine with Anaconda IDE installed.

8 Future Scope and further enhancement of the Project

It is easier to extend the Model that we have developed to similar or other related variety of problems and Business Applications. We could enhance the train dataset or the entire dataset for that matter to provide a much more better model, which means our model could work smoothly on enormously large datasets with the similar problem i.e. Titanic Survivors Dataset to

get better insights on doubtful data and could predict over forthcoming data using attributes learnt by this model. The same modelling procedure, the entire process along with the Algorithm can be also applied to other relevant and closely related Business Problems like the Financial Agents Prediction for a Sales Company, Predicting Defective Items in a Production Line, Credit Card Defaulters Prediction, Customer Churn Prediction for Financial Products like Savings Bank Account for a Bank etc. and so many more, the similarity being all are Binary Classification Problem and the model we created here suits the need and the best and the fastest Algorithm for such would be Logistic Regression, however, Ensemble Techniques are always recommended.

9 Project Repository Location

S#	Project Artifacts (softcopy)	Location (Mention Lab-ID, Server ID, Folder Name etc.)	Verified by Project Guide	Verified by Lab In-Charge
1.	Project Synopsis Report (Final Version)		Name and Signature	Name and Signature
2.	Project Progress updates		Name and Signature	Name and Signature
3.	Project Requirement specifications		Name and Signature	Name and Signature
4.	Project Report (Final Version)		Name and Signature	Name and Signature
5.	Test Repository		Name and Signature	Name and Signature
6.	Any other document, give details		Name and Signature	Name and Signature

10 Definitions, Acronyms, and Abbreviations

Abbreviation	Description
DFD	Data Flow Diagram
IDE	Integrated Development Environment
OS	Operating system
RAM	Random Access Memory

11 Conclusion

The Titanic Survivors Prediction project developed is a Binary Classification Problem solved using Machine Learning Libraries of Python fully meets the objectives of the need for which it has been developed. The Model has reached a steady state where all possible identified and tested bugs have been eliminated. All sorts of Exploratory Analysis has been performed on the dataset prior to modelling and it was found that 342 passengers survived out of 891 on the ship, and the model was finally built using Logistic Regression Algorithm after rigorous testing using other techniques but LR was chosen as it performed better and was effective specific to this problem, finally I achieved approx. 74% accuracy on both train and test sets thus coming up as the Best Fitting Model, As so much as perfection is bothered no model is perfect ever and there square measure probabilities of improvement every moment. And, it could serve all similar variety of Binary Classification problems in the Industry provided it's retained on new data.

12 References

a) Books

- Advanced Engineering Mathematics by Erwin Keyszig
- Higher Engineering Mathematics by BS Grewal and HK Das
- Linear Algebra by Gilbert Stang and Think Stats by Allen B Downey
- Software Engineering Principles and Practices by Roger Pressman

b) Websites

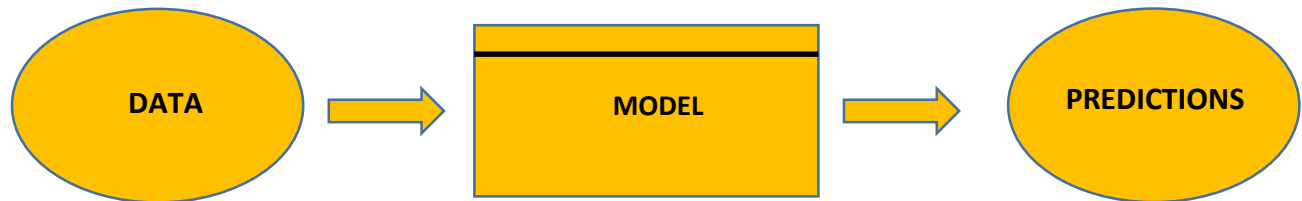
- www.google.com
- www.trainings.internshala.com
- www.youtube.com
- www.edx.com
- www.kaggle.com
- www.stackoverflow.com
- www.geeksforgeeks.com

S#	Reference Details	Owner	Version	Date
1.	Project Synopsis	Manish Gupta	1.0	03-12-20

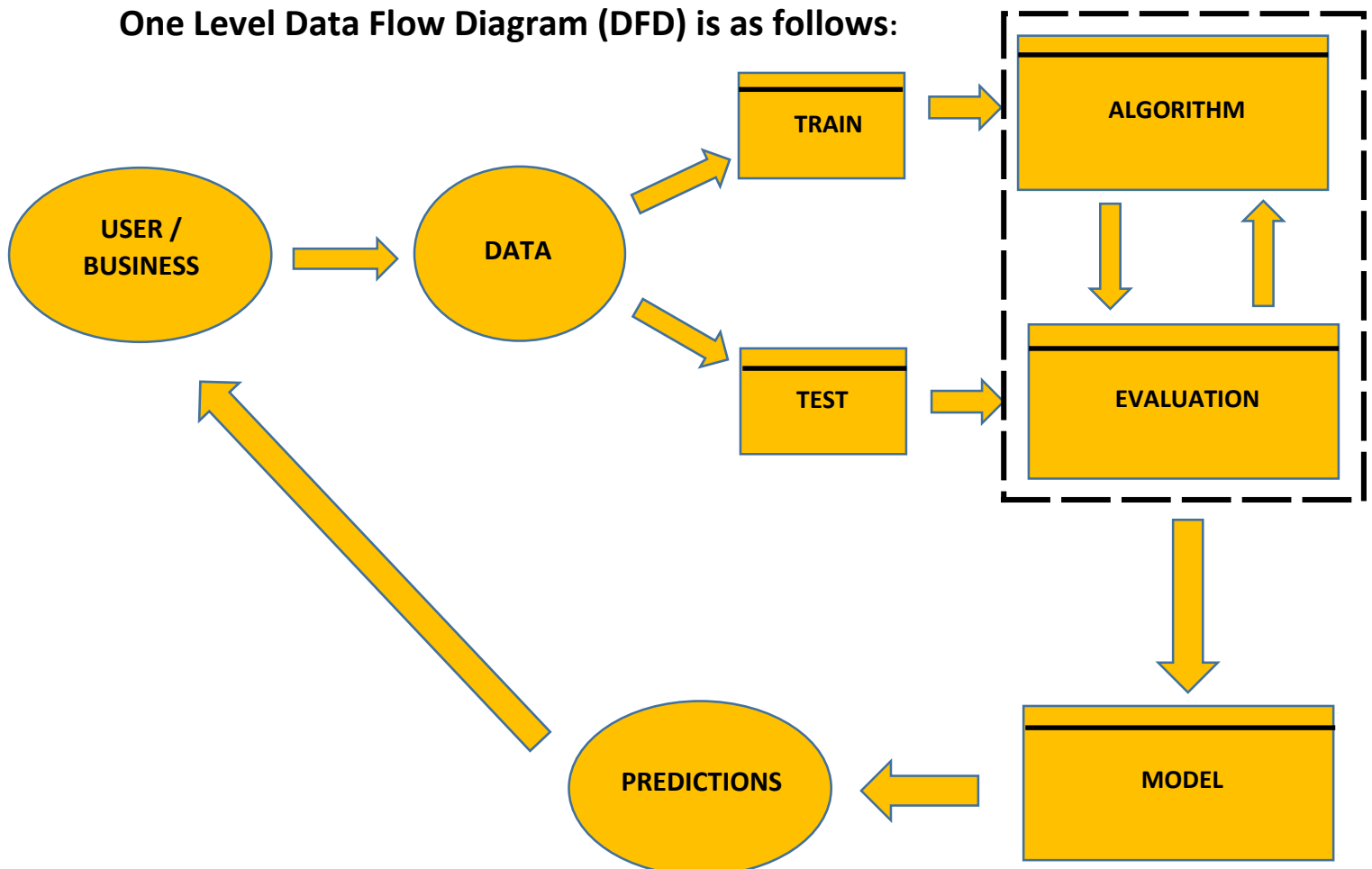
Annexure A

Data Flow Diagram (DFD)

Zero Level Data Flow Diagram (DFD) is as follows:



One Level Data Flow Diagram (DFD) is as follows:



Annexure B

Data Dictionary

Input Dataset

Variable	Definition	Key
PassengerId	Passenger ID	
Survived	Survival	0 = No, 1 = Yes
Pclass	Ticket class	1 = 1st, 2 = 2nd, 3 = 3rd
Sex	Sex	
Age	Age in years	
SibSp	# of siblings / spouses aboard the Titanic	
Parch	# of parents / children aboard the Titanic	
Ticket	Ticket number	
Fare	Passenger fare	
Cabin	Cabin number	
Embarked	Port of Embarkation	C = Cherbourg, Q = Queenstown, S = Southampton

Output Dataset

Variable	Definition	Key
PassengerId	Passenger ID	
Survived	Survival	0 = No, 1 = Yes

Annexure C

Screenshots

Launchpad – Home of Model Notebook:

Titanic Survivors Prediction

Problem Statement and Data Description

Titanic Survivors Prediction

The sinking of the Titanic is one of the most infamous shipwrecks in history.

While there was some element of luck involved in surviving, it seems some groups of people were more likely to survive than others.

In this challenge, our task as a Data Scientist is to build a predictive model that answers the question: "what sorts of people were more likely to survive?" using passenger data (ie name, age, gender, socio-economic class, etc).

Data Dictionary

There are multiple variables in the dataset as follows:

- Survival: Survival 0 = No, 1 = Yes
- pclass: Ticket class 1 = 1st, 2 = 2nd, 3 = 3rd
- sex: Sex
- Age: Age in years
- sibsp: # of siblings / spouses aboard the Titanic

Loading and Setting up the Dataset:

Loading the Dataset

```
In [2]: #importing necessary Library
import pandas as pd
#for plotting curves, can't work without this
%matplotlib inline
import matplotlib.pyplot as plt
```

```
In [3]: #initializing dataframe from pandas reference to dataset file Location
df = pd.read_csv(r"C:\Users\Manish Gupta\Desktop\ML Training\3. ML Life Cycle\Module 3\data.csv")
```

```
In [4]: #Checking Dimensions of added database
df.shape
```

```
Out[4]: (891, 12)
```

```
In [5]: #viewing top 5 rows in dataset
df.head()
```

```
Out[5]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S

Analysis and Exploration of the Dataset:

```
In [11]: #checking types of data in columns, object is parent of string, used for classification data, while int and float for continuous
df.dtypes
```

```
Out[11]: PassengerId    int64
Survived              int64
Pclass                int64
Name                  object
Sex                   object
Age                   float64
SibSp                 int64
Parch                 int64
Ticket                object
Fare                  float64
Cabin                 object
Embarked              object
dtype: object
```

```
In [12]: #Univariate Analysis - Tabular method on given dataset, central tendencies mentioned
df.describe()
```

```
Out[12]:
```

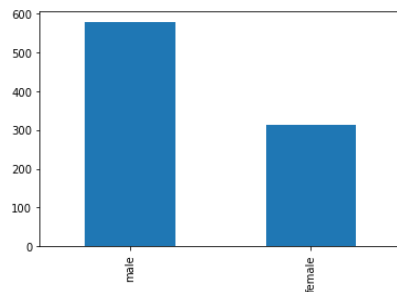
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000

```
In [23]: #Univariate Analysis for Categorical Data, here Sex for instance, using value_counts fun, tabular format
df['Sex'].value_counts()
```

```
Out[23]: male      577
female    314
Name: Sex, dtype: int64
```

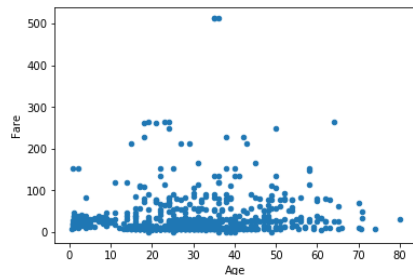
```
In [24]: #Univariate Analysis for Categorical Data, here Sex for instance, using value_counts fun, graphical format
df['Sex'].value_counts().plot.bar()
```

```
Out[24]: <matplotlib.axes._subplots.AxesSubplot at 0x253f572c2b0>
```



```
In [29]: #BVA Type1 - Continuous - Cont. BV Analysis, using Age and Fare Variables here, Graphical, using Scatter Plot
df.plot.scatter('Age', 'Fare')
```

```
Out[29]: <matplotlib.axes._subplots.AxesSubplot at 0x253f57dd518>
```



```
In [31]: #Checking Correlation between various Attributes in Dataset, 2D-Matrix Format Dataset with r values Corr. Coeff.
df.corr()
```

```
Out[31]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
PassengerId	1.000000	-0.005007	-0.035144	0.036847	-0.057527	-0.001652	0.012658
Survived	-0.005007	1.000000	-0.338481	-0.077221	-0.035322	0.081629	0.257307

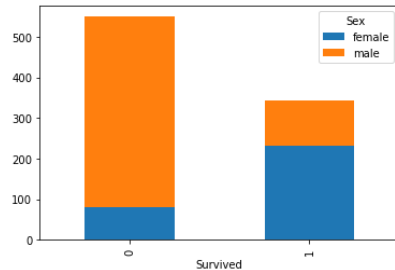
```
In [11]: pd.crosstab(df['Survived'], df['Sex'])
```

```
Out[11]:
```

Sex	female	male
Survived		
0	81	468
1	233	109

```
In [13]: pd.crosstab(df['Survived'], df['Sex']).plot(kind='bar', stacked=True)
```

```
Out[13]: <matplotlib.axes._subplots.AxesSubplot at 0x1b34dc0da90>
```



Cleaning Dataset and Building Model using Logistic Regression Algorithm:

```
Out[19]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare	Sex_female	Sex_male	Embarked_C	Embarked_Q	Embarked_S
0	1	0	3	22.0	1	0	7.2500	0	1	0	0	1
1	2	1	1	38.0	1	0	71.2833	1	0	1	0	0
2	3	1	3	26.0	0	0	7.9250	1	0	0	0	1
3	4	1	1	35.0	1	0	53.1000	1	0	0	0	1
4	5	0	3	35.0	0	0	8.0500	0	1	0	0	1

Segregating variables: Independent and Dependent Variables

```
In [20]: #seperating independent and dependent variables
x = data_c.drop(['Survived'], axis=1)
y = data_c['Survived']
x.shape, y.shape
```

```
Out[20]: ((891, 11), (891,))
```

Splitting the data into train set and the test set

```
In [21]: # Importing the train test split function
from sklearn.model_selection import train_test_split
train_x, test_x, train_y, test_y = train_test_split(x, y, random_state = 2)
```

Implementing Logistic Regression

```
In [26]: #importing Logistic Regression and metric F1-score
from sklearn.linear_model import LogisticRegression as LogReg
from sklearn.metrics import f1_score
```

```
In [27]: # Creating instance of Logistic Regression
logreg = LogReg()

# Fitting the model
logreg.fit(train_x, train_y)
```

```
Out[27]: LogisticRegression(C=1.0, class_weight=None, dual=False, fit_intercept=True,
intercept_scaling=1, max_iter=100, multi_class='ovr', n_jobs=1,
penalty='l2', random_state=None, solver='liblinear', tol=0.0001,
verbose=0, warm_start=False)
```

Making predictions using *predict* function

```
In [28]: # Predicting over the Train
train_predict = logreg.predict(train_x)
train_predict
```

Checking Model Accuracy and Exporting Predictions Dataset:

```
In [29]: # Calculating f1-score
k = f1_score(train_predict, train_y)
print('Training f1_score', k )

Training f1_score 0.734439834025

In [30]: # Predicting over the Test Set and f1-score
test_predict = logreg.predict(test_x)
k = f1_score(test_predict, test_y)
print('Test f1_score ', k )

Test f1_score      0.719512195122

In [31]: #Exporting the Predicted Results as as CSV Data file
ids = test_x['PassengerId']
survivors = logreg.predict(test_x)

#set the output as a dataframe and convert to csv file named survived.csv
output = pd.DataFrame({'PassengerId': ids, 'Survived': survivors })
output.to_csv('C:/Users/Manish Gupta/Desktop/survived.csv', index=False)

In [32]: #Viewing top 5 rows of the Predictions dataset
output.head()
```

Out[32]:

	PassengerId	Survived
707	708	0
37	38	0

End of Project Report



**Thank
You!!!**